

Chapter 1. Introduction

1.1. Background

The “Home Automation” concept has existed for many years. The terms “Smart home”, “Intelligent Home” followed and has been used to introduce the concept of networking appliances and devices in the house.

Home automation Systems (HASs) represents a great research opportunity in creating new fields in engineering, and computing. HASs includes centralized control of lighting, appliances, security locks of gates and doors and other systems, to provide improved comfort, energy efficiency and security system. HASs becoming popular nowadays and enter quickly in this emerging market. However, end users, especially the disabled and elderly due to their complexity and cost, do not always accept these systems.

Due to the advancement of wireless technology, there are several different of connections are introduced such as GSM, WIFI, and Bluetooth. Each of the connection has their own unique specifications and applications. Among the four popular wireless connections that often implemented in HAS project, WIFI is being chosen with its suitable capability. The capabilities of WIFI are more than enough to be implemented in the design. Also, most of the current laptop/notebook or Smartphone come with built-in WIFI adapter. It will indirectly reduce the cost of this system.

1.2 Problem Statement

The main objective of this program is to design a system that allows users upon authentication to remotely control and monitor multiple home appliances using a cell phone based interface. This service will be powerful and flexible tool that will offer thus service at any time and from anywhere with the constraints of the technologies being applied.

1.3. Project Objectives

- Android controlled Smart Home Automation should be able to control the home appliances wirelessly with effectively and efficiently.

1.4. Scopes

The project aims at designing a prototype for controlling the home appliances that can be controlled wirelessly via an application that provides the features of speech recognition, video streaming, and switch mode. An application is run on android device. The system can be used in wide range of areas.

The system integrated with different features can be applied in the following fields.

- The system can be used in home small offices to the big malls.
The system can be used from home to offices to control the electrical appliances.
- For remote access of appliances in internet or intranet
- For the development of technology friendly environment
The system incorporates the use of technology and making smart home automation. By the use of day to day gadgets we can utilize them for different prospective.

1.5. Application of project:

1. Automatic air cooling fan based on temperature.
2. Light control from cellphone from anywhere.
3. Monitoring water tank and auto filling it without having to go to the tank by ourself.

CHAPTER 2

LITERATURE REVIEW

Automation performs increasingly vitro role in daily experience and global economy. Engineers strive to combine automated device with mathematical and organizational tools to create complex system for a rapidly expanding range of application and human activities.

A group of people implemented internet for things for home automation. They presented a paper on the implementation of Internet of things for home automation. This paper mainly focused on IoT coverage that connects all the variety of objects like smart phone, tablets, digital cameras and sensors in the internet and thus provides many services and huge amount of data and information. They also focused on Cloud computing, Cloud based platform help to connect the things that surrounds as so that we can easily access anything at any time and in any place. They have illustrated sensing as a service on cloud by using certain application like Augmented Reality, Agriculture, Environment monitoring etc. and finally they have proposed a prototype model for providing sensing as a service on cloud. The society need new and scalable, compatible and secure solutions for both the management of the ever broader complexly networked Internet of Things. Security concern is overcome by this model since we are using Wi-Fi Wireless Equivalent Privacy (WEP) and Wi-Fi Protected Access (WPA) are two most used security accesses used in Wi-Fi [1].

Bluetooth based wireless home automation system using fpga: They presented a paper on the Bluetooth based Wireless Home automation system using FPGA. They primarily focused on Bluetooth technology. With the help of the Bluetooth module (HC-05) and Android Phone, they control the home appliances, which all connected to FPGA board. Thus, they have mentioned the advantages of the home automation, which not only reduces the human efforts, but it is also energy efficient and time saving. Moreover, they have included that it is also help to the handicapped and old aged people to control the home appliance without any difficulties. We need module so that the range will be high as well as it can operate in different frequencies. This drawback is overcome by our model. Wi-Fi based networks work at 2.4, 3.6 and 5 GHz. In addition, it can extend up to range 100m [2].

Hand gesture based home automation for visually challenged: They published a paper entitled Hand gesture based Home Automation for Visually Challenged People. They have designed a device for the visually challenged people to help them to operating the home appliances. They have used MEMS (Microelectromechanical Systems) accelerometer which is used to sense the accelerations of a hand in corresponding three perpendicular direction that is (x y z) and thus transmit the signal to wireless protocol using Radio frequency. The gesture templates were stored in a microcontroller at the receiver end. The received gesture and the hand gesture were compared by the templates. If the corresponding gesture were matched with the templates then accordingly home appliances were controlled. In addition, these devices were help for the old aged person too. Since they have used four types of gesture and stored in the microcontroller and it processed further. However, we need the system to be automated without the use of gesture. We do not need the gesture to be stored in the controller. Nowadays, the application can be used by any means of people by the option "TALKBACK" in the android application [3].

Home automation using atmega328 microcontroller and android application: They presented a paper on Home automation using AT mega Microcontroller and Android application. In this paper they have describe the design and development of a remote household appliance control system using the ATmega328 microcontroller and android mobile through GSM technology. In addition, this appliances remotely using the SMS-based system that satisfying user needs and requirements. Thus, all electrical household appliances can be controlled by sending a text message from an Android mobile. For Controlling, the remote appliances carried out by sending a SMS message from a mobile phone, which again congestion process and make system, complicated for the disabled persons. Here, we does not carry out this technique, we are using simple open source android application through Wi-Fi we can directly control the entire appliance with a greater extent. Thus, intend to be a reliable method [4].

E-mail interactive home automation system: They presented a paper on E-mail interactive Home automation system. They have enlighten on a basic home automation application on the public domain through the subject of E-mail ID. The switching action were done by LED indication. They provide a basic application of home automation using GVT app, which can be easily implemented and used as efficiently. The coding which they provide is generic and flexible in user-friendly manner and can be controlled in any application like power control, surveillance etc easily. In addition, all the results were generated by a series of E-mail sent to the user of G-mail account. For each and every interrupts one email will generated and will send to the user of the Gmail account, which again a tedious process. We can easily control the appliances instead of going through such process [5].

The main objective of another paper is to develop a home automation system using an Arduino board with Bluetooth being remotely controlled by any Android OS smart phone. As technology is advancing so houses are also getting smarter. Modern houses are gradually shifting from conventional switches to centralized control system, involving remote controlled switches. Presently, conventional wall switches located in different parts of the house makes it difficult for the user to go near them to operate. Even more it becomes more difficult for the elderly or physically handicapped people to do so. Remote controlled home automation system provides a most modern solution with smart phones. In order to achieve this, a Bluetooth module is interfaced to the Arduino board (ATMEGA 328) at the receiver end while on the transmitter end, a GUI application on the cell phone sends ON/OFF commands to the receiver where loads are connected [6].

Another paper presents the design of the low cost voice recognition based home automation system for the physically challenged people suffering from quadriplegia or paraplegia (who cannot move their limbs but can speak and listen) to control the various home appliances and can actuate the bed elevation just by the voice commands according to their need and comfort. The proposed system consists of a voice recognition module, Arduino uno microcontroller, relay circuit to and an adjustable bed. The voice recognition module needs to be trained first before it can be used to recognize commands. Upon successful recognition of voice command the Arduino drives the corresponding load with the help of the relay circuit. The adjustable bed elevation can be set to the three different modes as per the user comfort and need. The accuracy of voice recognition module is also measured in different conditions. The experimental results validate the functions of the proposed system. The results show the system can provide great assistant to the physically challenged people without any third person's assistances [7].

Chapter 3

Related Theory

3.1 Arduino Uno:

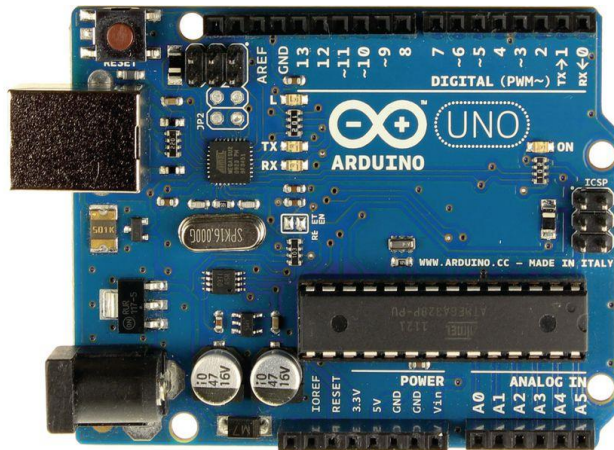


Fig 3.1: Arduino UNO

The **Arduino Uno** is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable.

The Arduino project started at the Interaction Design Institute Ivrea (IDII) in Ivrea, Italy. At that time, the students used a BASIC Stamp microcontroller, at a cost that was a considerable expense for many students. In 2003, Hernando Barragán created the development platform Wiring as a Master's thesis project at IDII, under the supervision of Massimo Banzi and Casey Reas, who are known for work on the Processing language. The project goal was to create simple, low-cost tools for creating digital projects by non-engineers. The Wiring platform consisted of a printed circuit board (PCB) with an ATmega168 microcontroller, an IDE based on Processing, and library functions to easily program the microcontroller. In 2003, Massimo Banzi, with David Mellis, another IDII student, and David Cuatrecasas, added support for the cheaper ATmega8 microcontroller to Wiring. But instead of continuing the work on Wiring, they forked the project and renamed it Arduino. Early arduino boards used the FTDI USB-to-serial driver chip and an ATmega168. The Uno differed from all preceding boards by featuring the ATmega328P microcontroller and an ATmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter.

Each of the 14 digital pins and 6 analog pins on the Uno can be used as an input or output, under software control using pin Mode (), digital Write (), and digital Read () functions. They operate at 5 volts. Each pin can provide or receive 20 mA as the recommended operating condition and has an internal pull-up resistor (disconnected by default) of 20-50K ohm. A maximum of 40mA must not be exceeded on any I/O pin to avoid permanent damage to the microcontroller. The Uno has 6 analog inputs, labeled

A0 through A5; each provides 10 bits of resolution (i.e. 1024 different values). By default, they measure from ground to 5 volts, though it is possible to change the upper end of the range using the AREF pin and the analog Reference () function.

In addition, some pins have specialized functions:

- **Serial / UAR:** pins 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL serial chip.
- **External interrupts:** pins 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.
- **PWM** (pulse-width modulation): pins 3, 5, 6, 9, 10, and 11. Can provide 8-bit PWM output with the analog Write () function.
- **SPI** (Serial Peripheral Interface): pins 10 (SS), 11 (MOSI), 12 (MISO), and 13 (SCK). These pins support SPI communication using the SPI library.
- **TWI** (two-wire interface) / I²C: pin SDA (A4) and pin SCL (A5). Support TWI communication using the Wire library.
- **AREF** (analog reference): Reference voltage for the analog inputs.

3.2 ESP8266 (Wi-Fi Module):

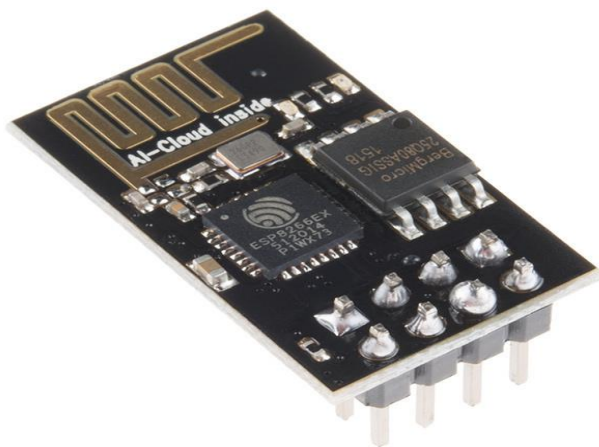


Fig 3.2 ESP8266

The **ESP8266** is a low-cost Wi-Fi microchip with full TCP/IP stack and microcontroller capability produced by manufacturer Espressif Systems in Shanghai, China. The chip first came to the attention of Western makers in August 2014 with the **ESP-01** module, made by a third-party manufacturer Ai-Thinker. This small module allows microcontrollers to connect to a Wi-Fi network and make simple TCP/IP connections using Hayes-style commands. However, at first there was almost no English-language documentation on the chip and the commands it accepted.^[2] The very low price and the fact that there were very few external components on the module, which suggested that it could eventually be very inexpensive in volume, attracted many hackers to explore the module, chip, and the software on it, as well as to translate the Chinese documentation.^[3]

The **ESP8285** is an ESP8266 with 1 MiB of built-in flash, allowing for single-chip devices capable of connecting to Wi-Fi. The successor to these microcontroller chips is the ESP32, released in 2016.

3.3 Breadboard & Jump Wire:

A breadboard is a construction base for prototyping of electronics. Originally the word referred to a literal bread board, a polished piece of wood used for slicing bread. In the 1970s the solderless breadboard (a.k.a. plug board, a terminal array board) became available and nowadays the term "breadboard" is commonly used to refer to these. Because the solderless breadboard does not require soldering, it is reusable. This makes it easy to use for creating temporary prototypes and experimenting with circuit design. For this reason, solderless breadboards are also popular with students and in technological education. Older breadboard types did not have this property. A stripboard (Vero board) and similar prototyping printed circuit boards, which are used to build semi-permanent soldered prototypes or one-offs, cannot easily be reused. A variety of electronic systems may be prototyped by using breadboards, from small analog and digital circuits to complete central processing units (CPUs).

Jump wires (also called jumper wires) for solderless bread boarding can be obtained in ready-to-use jump wire sets or can be manually manufactured. The latter can become tedious work for larger circuits. Ready-to-use jump wires come in different qualities, some even with tiny plugs attached to the wire ends. Jump wire material for ready-made or homemade wires should usually be 22 AWG (0.33 mm^2) solid copper, tin-plated wire - assuming no tiny plugs are to be attached to the wire ends. The wire ends should be stripped $\frac{3}{16}$ to $\frac{5}{16}$ in (4.8 to 7.9 mm). Shorter stripped wires might result in bad contact with the board's spring clips (insulation being caught in the springs). Longer stripped wires increase the likelihood of short-circuits on the board. Needle-nose pliers and tweezers are helpful when inserting or removing wires, particularly on crowded boards.



Fig 3.3: Jumper Wire

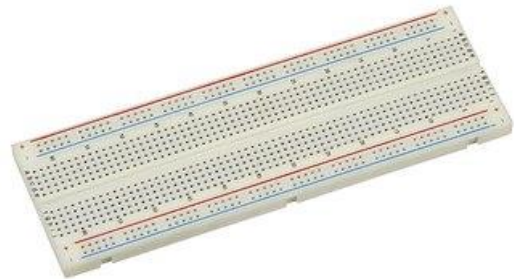


Fig 3.4: Breadboard

3.4 LM35 temperature sensor:

The LM35 series are precision integrated-circuit temperature sensors, whose output voltage is linearly proportional to the Celsius temperature. The LM35 operates at -55° to +120°C. The basic centigrade temperature sensor (+2°C to +150°C) is shown in figure below.

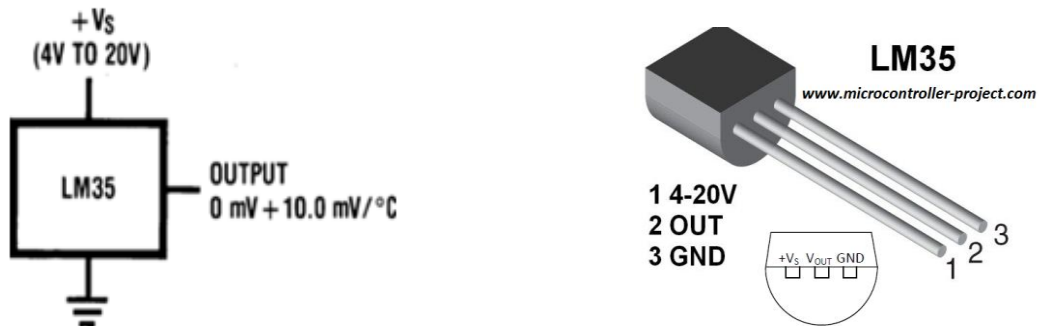


Fig 3.5 : LM35

Some features of this sensor are:

- Calibrated directly in $^\circ\text{Celsius}$ (Centigrade)
- Rated for full -55° to $+150^\circ\text{C}$ range
- Suitable for remote applications
- Low cost due to wafer-level trimming
- Operates from 4 to 30 volts
- Low self-heating,
- $\pm 1/4^\circ\text{C}$ of typical nonlinearity

Operation of LM35:

- The LM35 can be connected easily in the same way as other integrated circuit temperature sensors. It can be stuck or established to a surface and its temperature will be within around the range of 0.01°C of the surface temperature.
- This presumes that the ambient air temperature is just about the same as the surface temperature; if the air temperature were much higher or lower than the surface temperature, the actual temperature of the LM35 die would be at an intermediate temperature between the surface temperature and the air temperature.

The temperature sensors have well known applications in environmental and process control and also in test, measurement and communications. A digital temperature is a sensor, which provides 9-bit temperature readings. Digital temperature sensors offer excellent precise accuracy, these are designed to read from 0°C to 70°C and it is possible to achieve $\pm 0.5^\circ\text{C}$ accuracy. These sensors completely aligned with digital temperature readings in degree Celsius.

3.5 Liquid Crystal Display:

LCD is the display device which is of 16x2 size and it has yellow background light. This LCD is connected to Arduino. The following is the interfacing diagram of LCD with Arduino uno. Fig. LCD interfacing with Arduino the vehicle. The following figure shows the LCD display. To enable terminal latch of LCD high to low pulse is sent and RS bit is enabled. Once the latch is enabled the data is transferred through the interfacing pins parallel and the LCD shows the display on it. These LCD are easy to

program and they are economical too. LCD interfacing with microcontroller is very easy. Here in our project the level of water in the tank is displayed on the LCD.



Fig 3.6: LCD display

Chapter 4. Methodology

4.1 Block Diagram

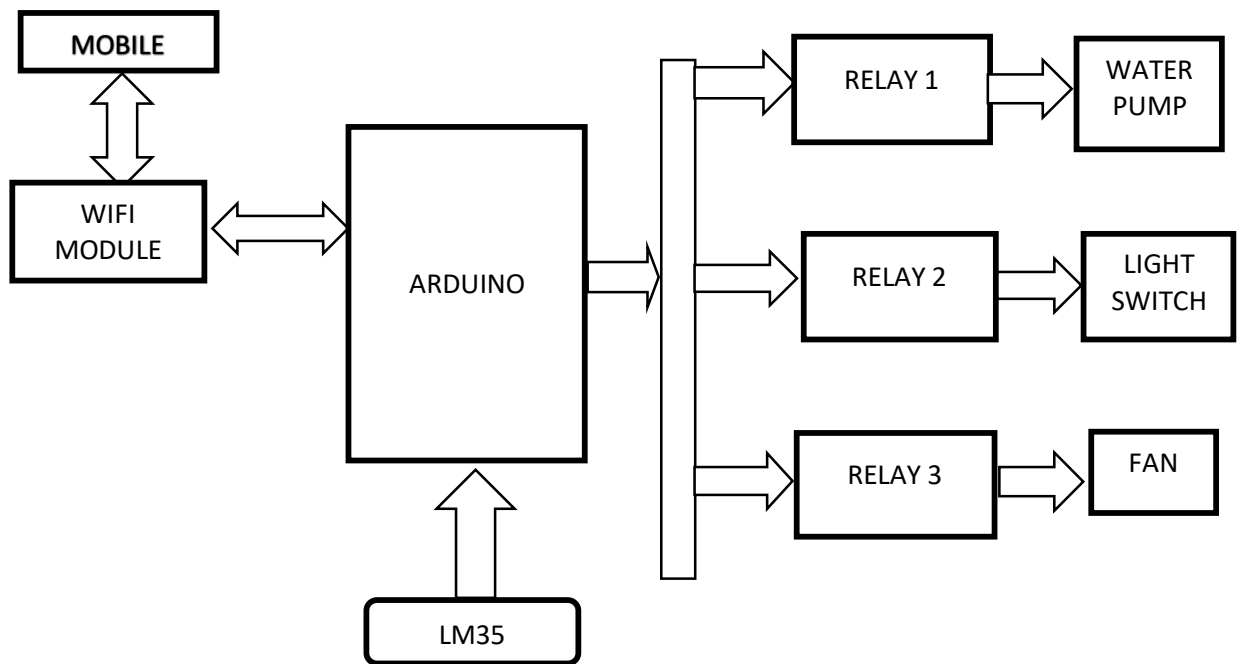


Fig: Block diagram of home automation system.

4.2 ALGORITHM:

For water level indicator:

Step 1: Start

Step 2: Sense water in sump tank

Step 3: Check if tank empty?

 If yes: buzzer on and goto step 1 and start filling tank with water

 If no: sense water level in tank

Step 4: Use of Arduino for decision making based on water level

Step 5: Check if tank full?

 If yes: motor off and goto step 1

 If no: motor on

Step 6: End

For temperature sensor:

Step 1: Start

Step 2: Sense temperature using sensor LM35

Step 3: Check if temp is equal to 25?

 If yes: Fan on

 If no: Fan off

Step 4: Check if temp is equal to 30?

 If yes: Fan on at 300rpm

 If no: Check if temp is equal to 35

 If yes: Fan on at 400rpm

Step 5: End

For Remote control light switch by mobile:

Step 1: Start

Step 2: Give command in from mobile

Step 3: Send to Arduino using Wi-Fi

Step 4: Decode the command in Arduino

Step 5: Use bulb accordingly

Step 6: Stop

4.3 FLOWCHART

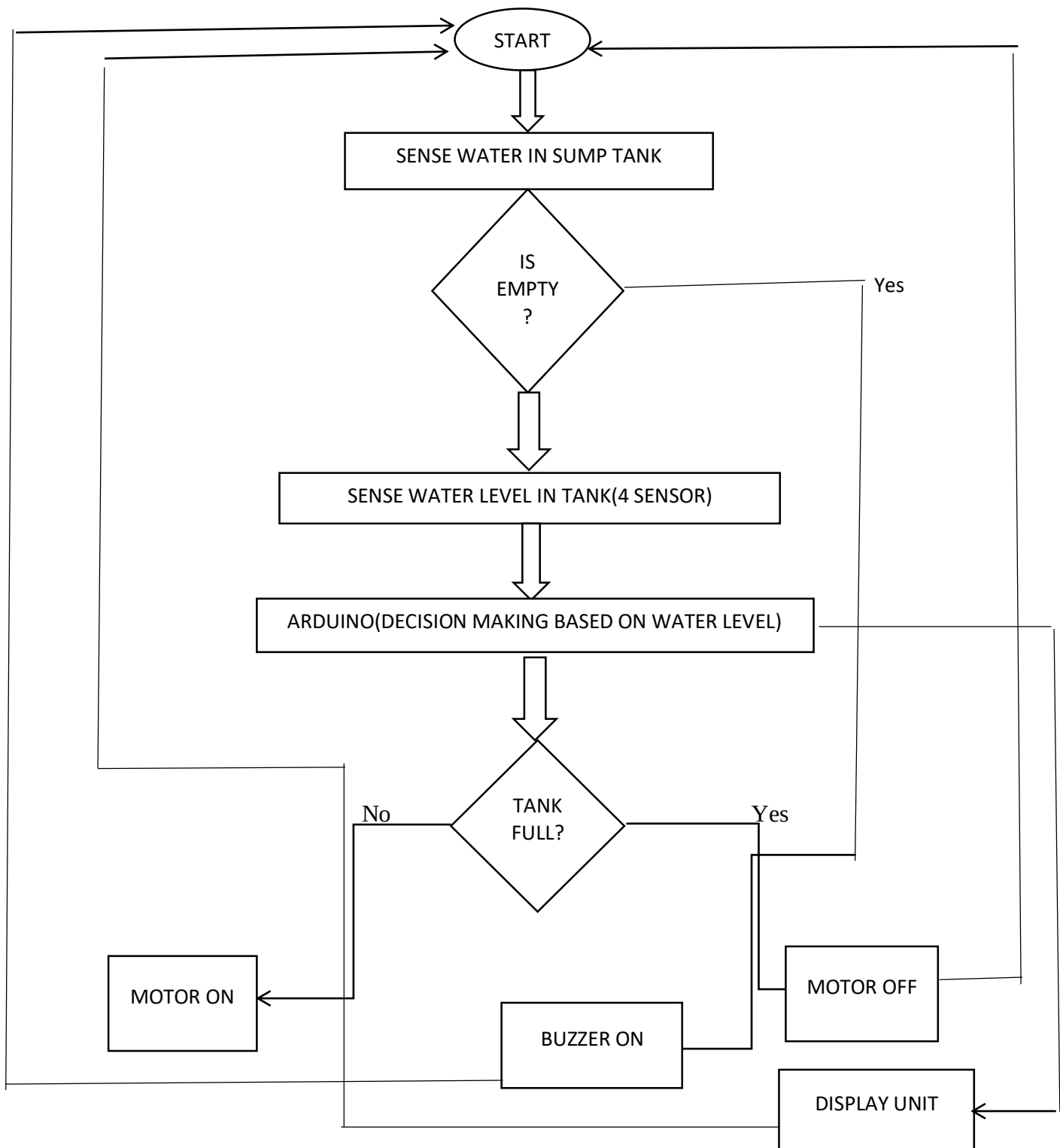


Fig :Flowchart of water level indicator and pump controller

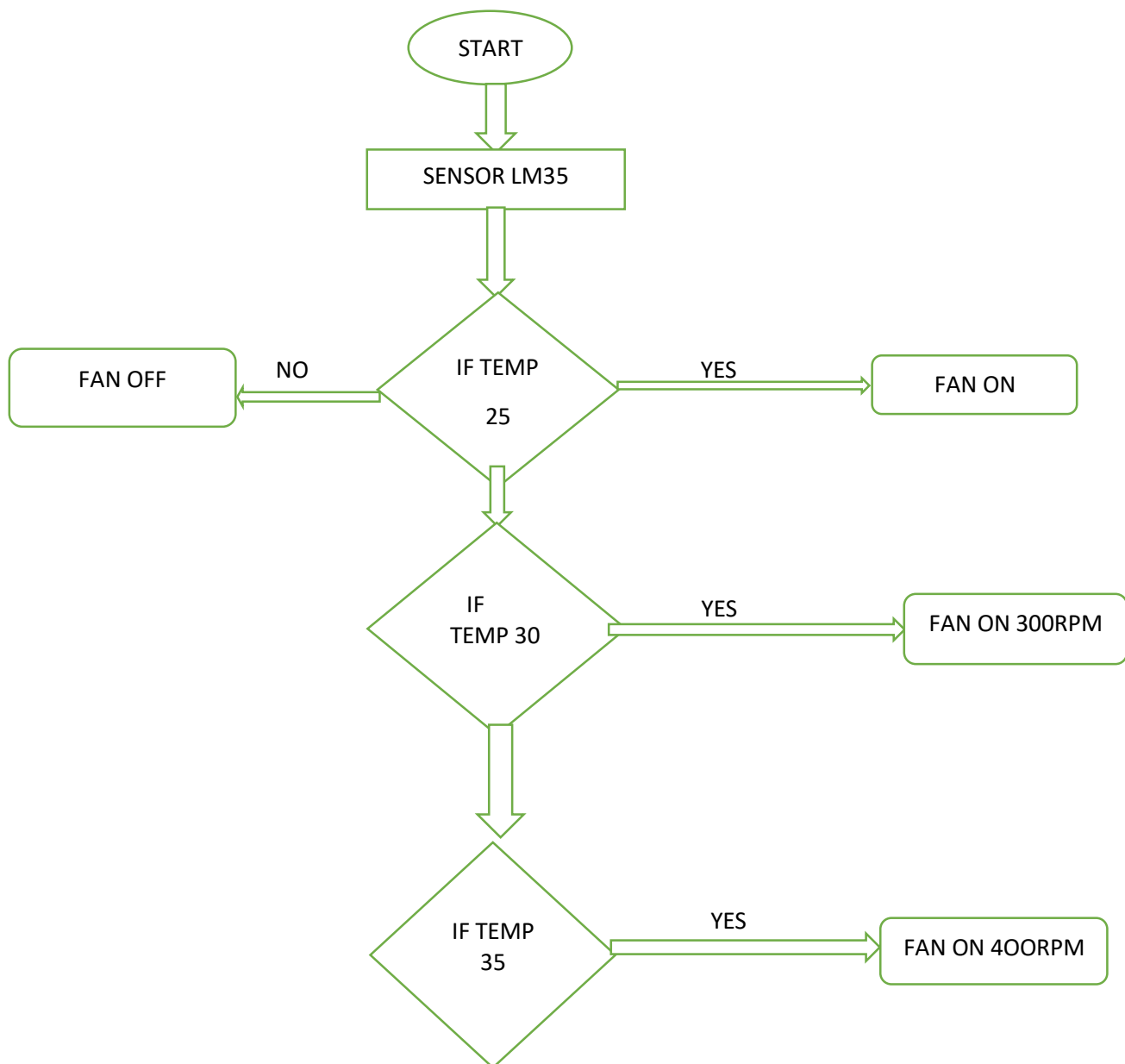


Fig: Flowchart of Temperature based Fanspeed Controller

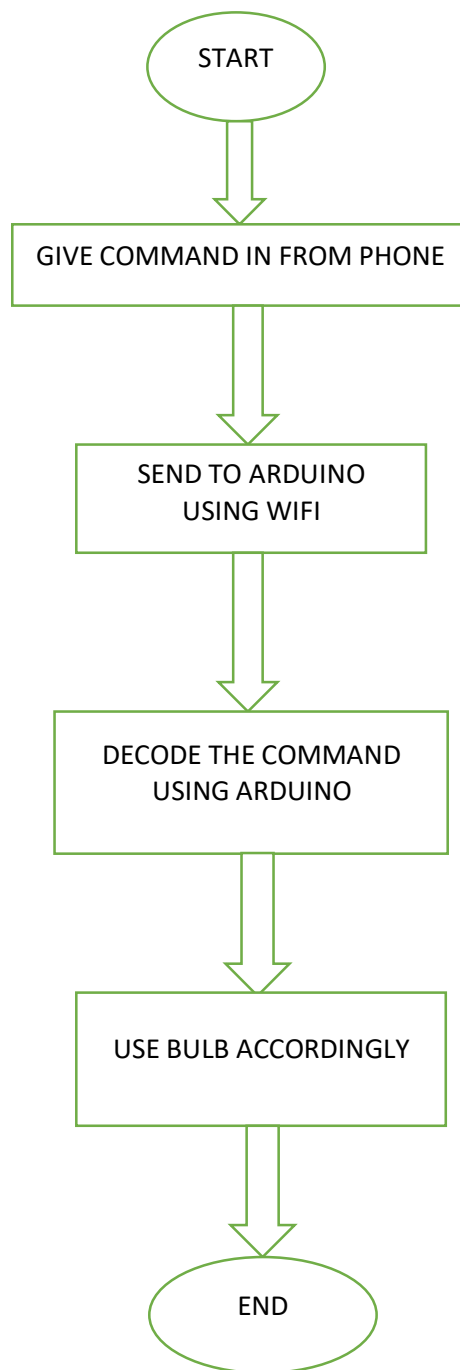


Fig: Flowchart for remote control light switch using mobile

CHAPTER 5

Expected Output

We made a complete smart home automation system and successfully implemented in real time.

In this project we build three different system that is

1. Automatic water level monitor and pump controller , which indicate the level of water if it is empty it fill the tank and when the water level is full it stop the flow of water and send signal to LCD “water level full”.
2. Temperature based fan speed control, Here we provide the reference temperature as 25degree Celsius that is room temperature when the temperature raise or increase above the reference temperature then the fan start rotating until particular room temperature is not below than reference temperature .
3. Controllable light switch ,here we give command to smart device either on or off accordingly light perform its task .

In automatic water level monitor and pump controller when the tank was full, it was sensed by resistor and this signal was sent to arduino which ultimately stopped the water pump from pumping further water to the tank.

In temperature based fan speed control, the temperature was sensed by LM35 and sent to arduino which controlled the speed of the fan accordingly.

In Controllable light switch the switch of the light was synced to the mobile with the help of Wi-Fi module and was successfully controlled through our mobile.

CHAPTER 6

Epilogue

6.1 Gantt Chart

Task	Duration							
	Semester-5				Semester-6			
	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul
Research								
Detail System Analysis								
Programming Hardware and Design								
Testing								
Final Product								
Documentation								

6.2 REFERENCE

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